AYE
TECH HUB

• HVAC ENGINEERING • LESSON 02

The Refrigeration Cycle
How Cooling Works

Compress, Condense, Expand, Evaporate — Repeat

Lesson 2 of 30 | HVAC Engineering Program | Beginner

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- The vapor compression cycle — 4 stages every HVAC engineer must know
- How each component works: compressor, condenser, EXV, evaporator
- Refrigerants — R-22, R-410A, R-32, and the low-GWP transition
- COP and EER — how to measure and compare system efficiency
- Heat pump operation — cooling and heating from one machine

01 | What Is the Refrigeration Cycle?

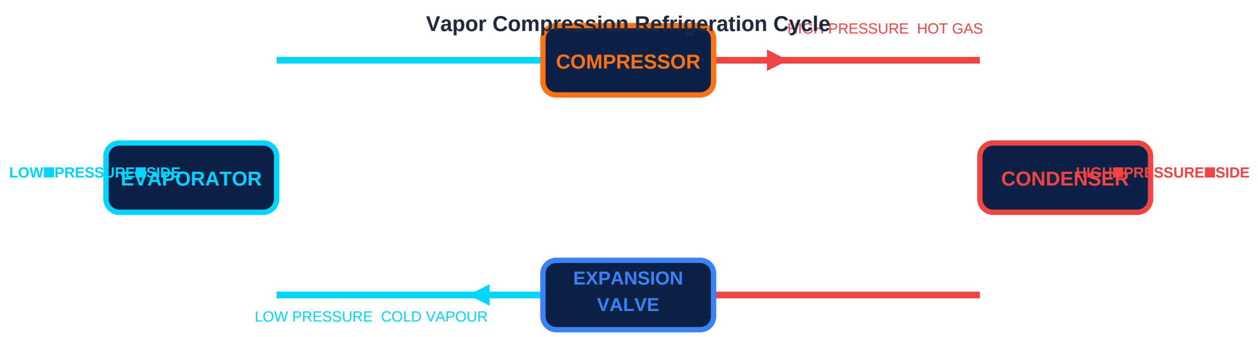
The refrigeration cycle is the thermodynamic process that **moves heat from a low-temperature space to a high-temperature environment**. This seems to defy nature — heat normally flows from hot to cold — but by adding mechanical work (the compressor), we can force heat to move in the opposite direction. This is how every air conditioner, refrigerator, heat pump, and chiller in the world operates.

The most common type is the **vapor compression refrigeration cycle**, invented by Jacob Perkins in 1834. It uses a **refrigerant** — a working fluid that repeatedly changes state between liquid and vapour — to absorb heat from one place and release it in another. Understanding this cycle is the single most important concept in HVAC engineering.

> **THE CORE PRINCIPLE — Moving Heat, Not Creating Cold**
>
> Air conditioning does not "create cold." It removes heat from inside a building and dumps it outside. A 5 kW air conditioner compressor uses 5 kW of electrical work to move perhaps 15 kW of heat — extracting 10 kW from indoors and rejecting 15 kW outdoors. The cold feeling is the absence of heat, not the presence of something cold.

02 | The 4-Stage Vapor Compression Cycle

The cycle has exactly four stages, each performed by a dedicated component. The refrigerant continuously circulates through these four stages, changing pressure and state as it goes.



Stage	Component	What Happens to Refrigerant	Pressure	Temperature	State Change
① Compression	Compressor	Low-pressure vapour is compressed to high pressure. Work is added — this is where electricity is consumed.	Low → High	Low → High	Vapour → Superheated Vapour
② Condensation	Condenser	Hot high-pressure vapour rejects heat to the outside air. Refrigerant cools and condenses into liquid.	High	High → Medium	Superheated Vapour → Liquid
③ Expansion	Expansion Valve (EXV/TXV)	High-pressure liquid passes through a small orifice and flashes to low pressure. Temperature drops sharply.	High → Low	Medium → Very Low	Liquid → Liquid-Vapour Mix
④ Evaporation	Evaporator	Cold low-pressure refrigerant absorbs heat from the conditioned space. It boils and becomes vapour again.	Low	Very Low → Low	Liquid Mix → Vapour

03 | The 4 Key Components — How Each One Works

Component	Function in the Cycle	Key Engineering Parameters	Common Types
Compressor	The "heart" of the system — compresses low-pressure refrigerant vapour to high pressure, driving the entire cycle	Capacity (kW/TR), displacement (cm³/rev), compression ratio, isentropic efficiency (70–85%)	Reciprocating, scroll (residential), screw (commercial), centrifugal (large chiller)
Condenser	Rejects heat from the high-pressure hot refrigerant to the outside environment — refrigerant condenses from vapour to liquid	Condensing temperature (°C), subcooling (typically 5–10°C), LMTD, UA value	Air-cooled (fans), water-cooled (cooling tower), evaporative condenser
Expansion Device	Reduces refrigerant pressure from high side to low side — causes temperature drop and partial flashing to vapour	Orifice size, superheat setting (typically 5–8°C), pressure differential	TXV (thermostatic), EXV (electronic — most modern), capillary tube (small systems), float valve
Evaporator	Absorbs heat from the conditioned space into the cold low-pressure refrigerant — air or water is cooled across the coil	Evaporating temperature (°C), superheat at outlet (5–8°C), coil face velocity, fin spacing	Fin-and-tube coil (DX), flooded shell-and-tube (chiller), plate heat exchanger

04 | Refrigerants — The Working Fluid of the Cycle

The refrigerant is the fluid that carries heat around the cycle. It must have specific thermodynamic properties: it must boil at low pressure at useful evaporating temperatures and condense at manageable pressures. Refrigerant choice has become a major industry concern due to **environmental regulations** — both ozone depletion potential (ODP) and global warming potential (GWP) are tightly regulated.

Common Refrigerants — GWP Comparison & Industry Status

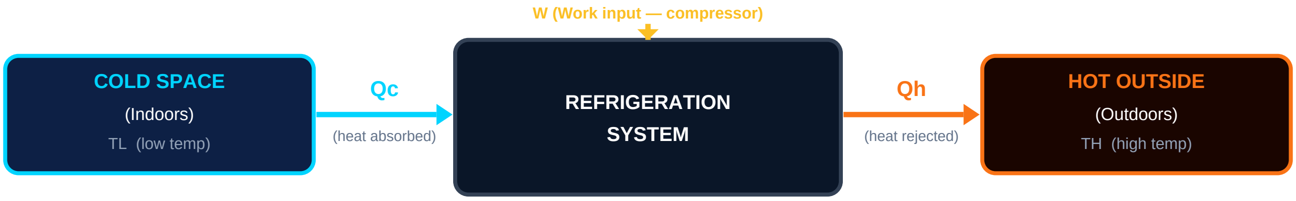
R-22	R-410A	R-32	R-454B
HCFC	HFC	HFC	HFO blend
GWP: 1,810	GWP: 2,088	GWP: 675	GWP: 466
Phased Out	Common Now	Transitional	Future Standard
(Montreal Protocol)	(phase-down ongoing)	(lower GWP)	(A2L — mildly flammable)

Refrigerant	Type	GWP	ODP	Normal Boiling Pt	Status & Notes
R-22 (Freon)	HCFC	1,810	0.055	–40.8°C	Phased out in developed countries. Still found in older equipment. Illegal for new installations in most countries.
R-410A	HFC	2,088	0	–51.6°C (blend)	Current dominant refrigerant for residential AC. Phase-down under Kigali Amendment by 2036. Zero ODP.
R-32	HFC	675	0	–51.7°C	Replacing R-410A in new residential equipment. Lower GWP, higher efficiency. Mildly flammable (A2L).
R-454B	HFO blend	466	0	–51.8°C (blend)	Next-generation replacement. Accepted by most manufacturers. Mildly flammable (A2L). Near drop-in for R-410A.
R-134a	HFC	1,430	0	–26.4°C	Used in automotive AC, chillers, cold storage. Being replaced by R-1234yf in vehicles (GWP = 4).

R-744 (CO■)	Natural	1	0	–78.4°C (sublimes)	Ultra-low GWP natural refrigerant. Used in supermarkets, heat pumps, industrial. Requires very high pressures.

05 | System Efficiency — COP and EER

A refrigeration system uses electricity (work) to move heat. Efficiency measures how effectively it does this. The two key metrics are **COP (Coefficient of Performance)** and **EER (Energy Efficiency Ratio)**. A higher value means the system delivers more cooling or heating for each unit of electricity consumed.



Energy Balance: Qh = Qc + W

Metric	Formula	Typical Value	What It Means
COP (Cooling)	COP = Qc ÷ W	2.5 – 4.5	For every 1 kW of electricity used, 2.5–4.5 kW of heat is removed from the space
COP (Heating — Heat Pump)	COP = Qh ÷ W	3.0 – 5.5	For every 1 kW of electricity, 3–5.5 kW of heat is delivered — better than resistance heating (COP = 1)
EER	EER = Qc (BTU/hr) ÷ W (W)	8 – 15 BTU/W	Common in US market; EER = COP × 3.412
SEER	Seasonal average EER	14 – 25+	EER averaged over the whole cooling season — accounts for part-load operation
SCOP / HSPF	Seasonal COP	3.5 – 5.5 (SCOP)	Heat pump seasonal performance — used in EU energy labelling (A++ to G rating)

06 | Heat Pump — One Machine for Both Heating and Cooling

A **heat pump** is a refrigeration system with a reversing valve that allows it to operate in either direction. In cooling mode, heat flows from indoors to outdoors (air conditioning). In heating mode, the cycle reverses: heat is extracted from outside air (even at –15°C) and delivered indoors. Heat pumps are now the dominant technology for building heating and cooling worldwide.

Mode	Evaporator Location	Condenser Location	Heat Direction	COP Range
Cooling Mode	Indoors (fan coil unit)	Outdoors (condensing unit)	Heat removed from inside → rejected outside	2.5 – 4.5 (typical split AC)
Heating Mode	Outdoors (extracts heat from cold air)	Indoors (delivers heat)	Heat extracted from cold outdoor air → delivered inside	3.0 – 5.5 (even at –15°C outdoor)

KEY INSIGHT — Why Heat Pumps Beat Gas Boilers

A gas boiler converts 1 kWh of gas into roughly 0.9 kWh of heat (90% efficiency). A heat pump uses 1 kWh of electricity to deliver 3–5 kWh of heat (300–500% "efficiency" — actually COP). Even accounting for the higher cost of electricity vs gas, heat pumps typically cut heating energy costs significantly — and produce zero on-site CO■ emissions.

Reflection:

> *"Your room air conditioner is rated at 1.5 kW electrical input and 5 kW cooling capacity. Calculate its COP. How much heat is it actually rejecting to the outdoor unit? If you reversed it into heating mode with a COP of 4.0, how much heating power would it deliver for the same 1.5 kW input?"*

Action checklist:

Next up — HVAC-003:
HVAC Components — compressors, coils, fans, air handling units, and how they connect into a complete system.

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